

MR. MOL RYNA

LARAVEL MVC PATTERN

LESSON

The background of the slide is split into two main visual areas. The top half features a white background with a thin, wavy, grey line that flows from the left side towards the right. The bottom half of the slide shows a close-up, slightly blurred image of a laptop keyboard, with keys like 'End', 'F4', and 'F5' visible. A semi-transparent dark blue rectangle is positioned on the right side of the slide, containing a list of five items.

01 WHAT IS MVC ?

02 CONTROLLER WITH CREATE
WITH CMD

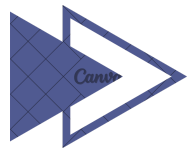
03 MODEL WITH MAKE WITH CMD

04 CONNECTIONG TO DB

05 MIGRATION WITH CREATE
MIGRATION

WHAT IS MVC ?

“MVC” stands for **Model – View – Controller**. It’s a software architectural pattern for organizing code in a way that separates concerns (data, presentation, logic). Laravel, as a PHP framework, uses the MVC pattern.



MVC PATTERN

MODEL

M

The Model is responsible for the **data layer**: it interacts with the database, represents business logic, handles the data structure. In Laravel, Models often use Eloquent ORM, which maps model classes to database tables.

View

V

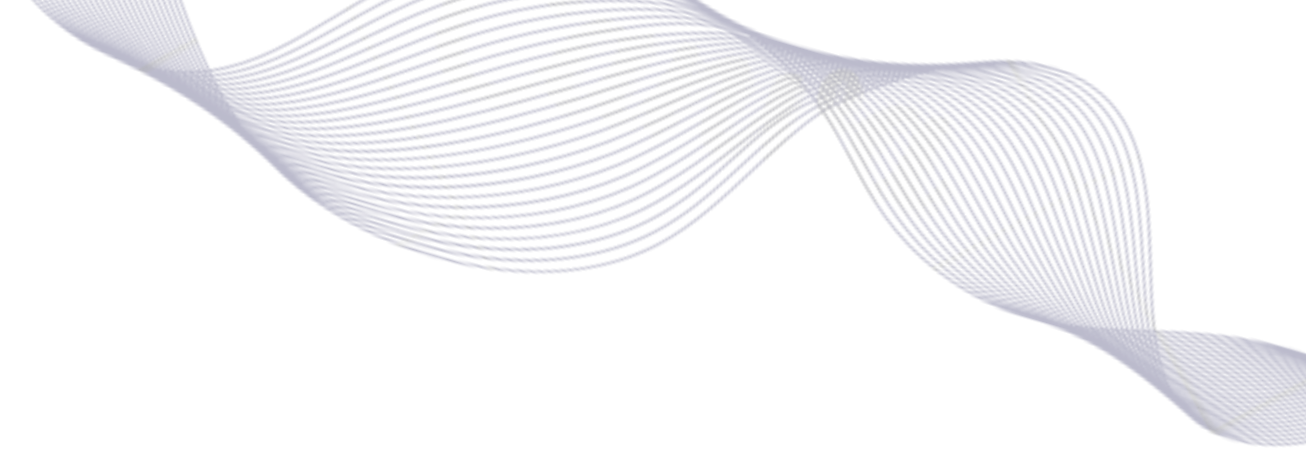
The View is the **presentation layer**: it displays data (usually HTML) to the user. In Laravel, views are typically Blade templates in resources/views

CONTROLLER

C

The Controller acts as a mediator between the Model and the View. It handles user requests (incoming HTTP), processes them (calls Models), and returns a Response (often selecting a View)





CONTROLLER

1. Basic Command to Create a Controller

COMMAND TO CREATE A CONTROLLER

```
php artisan make:controller ControllerName
```

EXAMPLE

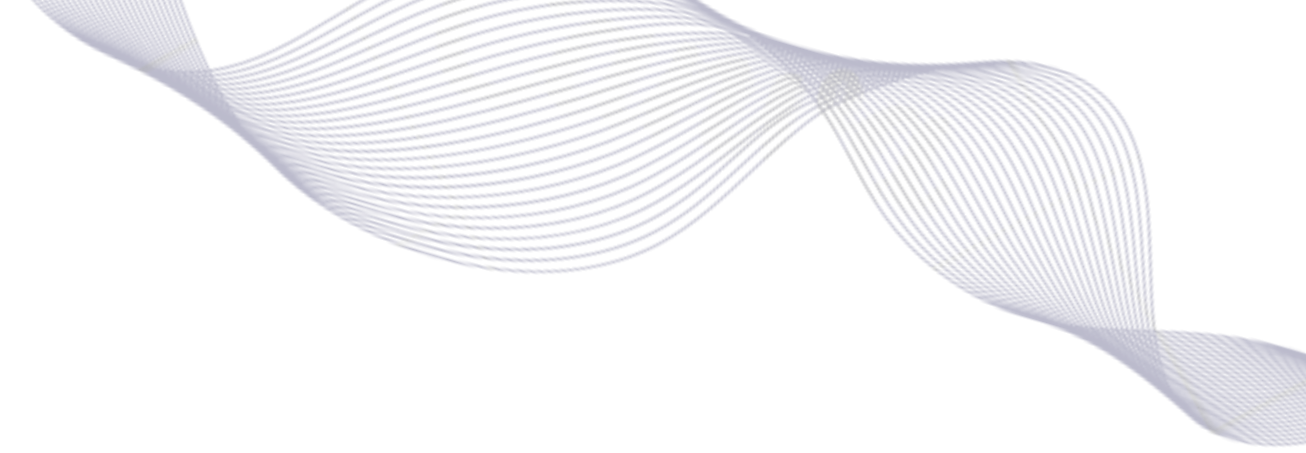
```
php artisan make:controller ProductController
```

RESULT

```
app/Http/Controllers/ProductController.php
```

After using this command you will see the result at **app/http/controller(path)**

ProductController



CONTROLLER

2. Create Controller in a Subfolder && Resource Methods

COMMAND TO CREATE A CONTROLLER

```
php artisan make:controller Folder/ControllerName
```

EXAMPLE

```
php artisan make:controller Admin/ProductController --resource
```

RESULT

```
app/Http/Controllers/Admin/ProductController.php
```

After using this command you will see the result at **app/http/controller(path)**

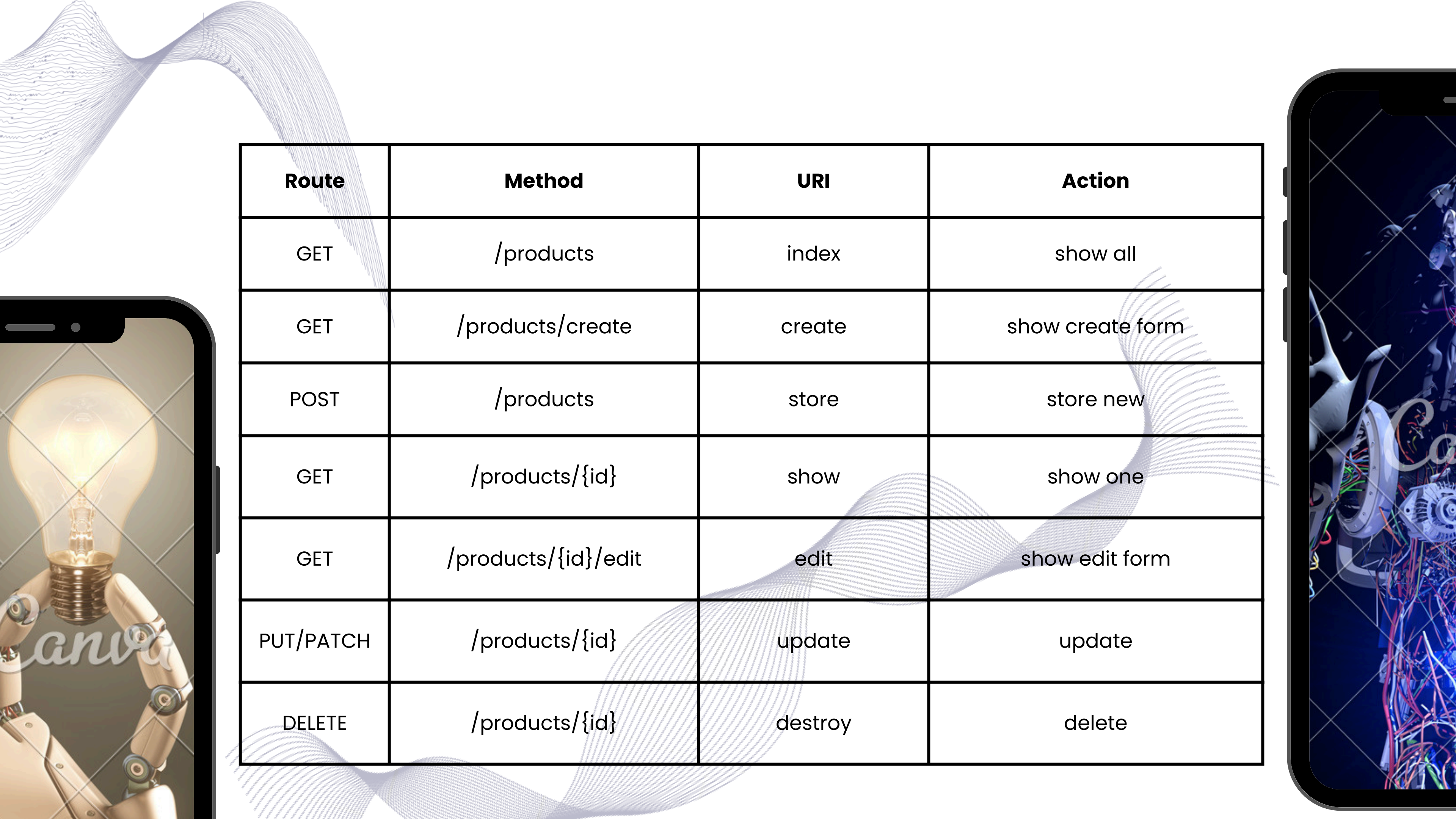
ProductController

EXAMPLE



ADD THIS TO YOUR ROUTES/WEB.PHP:

```
use App\Http\Controllers\ProductController;  
Route::resource('products',  
ProductController::class);
```

Route	Method	URI	Action
GET	/products	index	show all
GET	/products/create	create	show create form
POST	/products	store	store new
GET	/products/{id}	show	show one
GET	/products/{id}/edit	edit	show edit form
PUT/PATCH	/products/{id}	update	update
DELETE	/products/{id}	destroy	delete

BASIC COMMAND TO CREATE A MODEL

TERMINAL

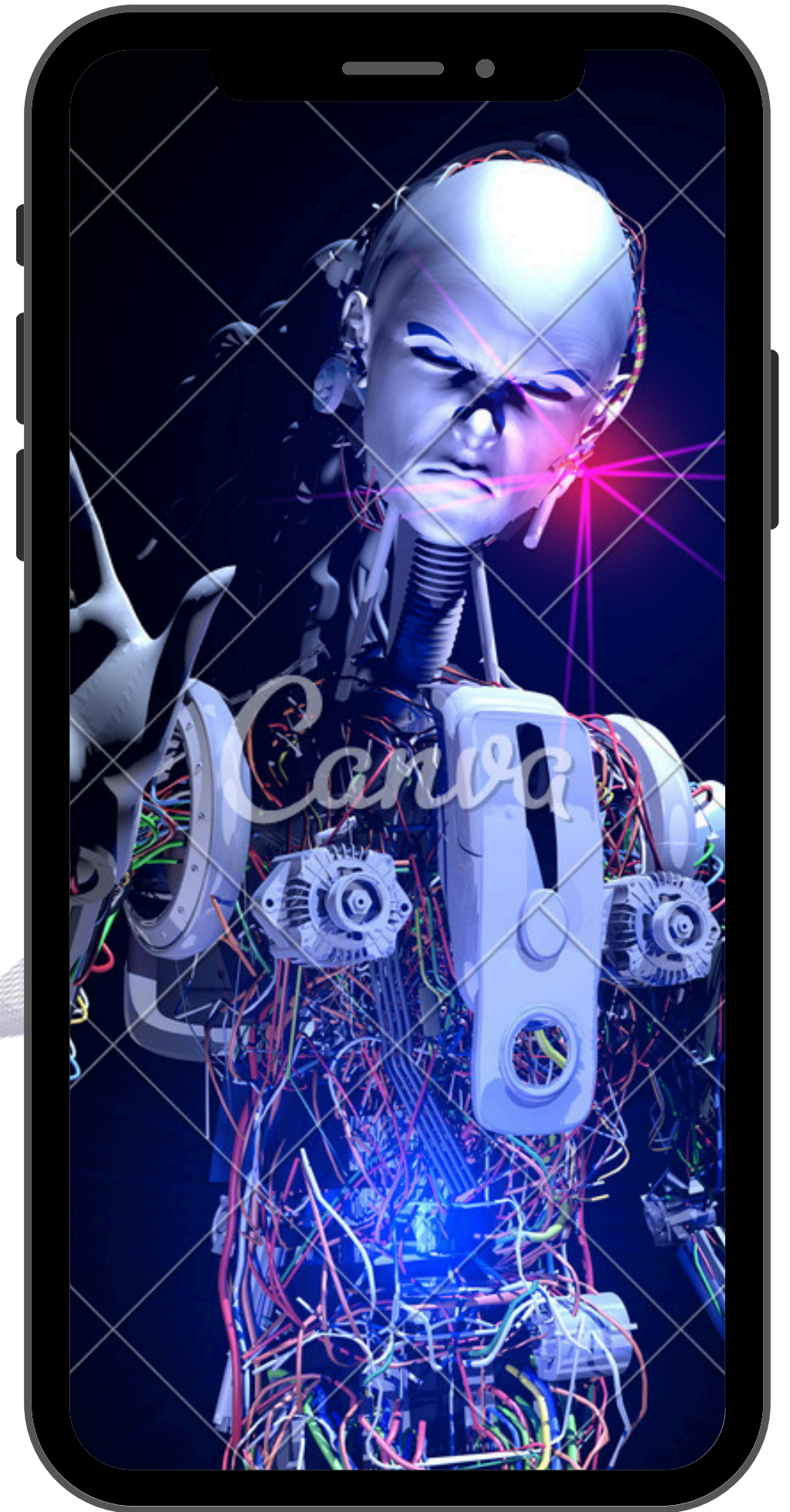
```
php artisan make:model ModelName
```

EXAMPLE

```
php artisan make:model Products
```

FOLDER

```
app/Models/Product.php
```



Inside the Model

```
<?php

namespace App\Models;

use
Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;

class Product extends Model
{
    use HasFactory;
}
```



EXAMPLE

EXAMPLE CREATE WITH MIGRATE

```
php artisan make:model Product --migration
```

RESULT

```
app/Models/Product.php  
database/migrations/2025_10_08_000000_create_products  
_table.php
```



Connect with database

.ENV FILE (ORIGINAL FILE)



```
DB_CONNECTION=sqlite
# DB_HOST=127.0.0.1
# DB_PORT=3306
# DB_DATABASE=
# DB_USERNAME=
# DB_PASSWORD=
```

.ENV FILE (AFTER CHANGED)



```
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE="learnlaravel"
DB_USERNAME="root"
DB_PASSWORD=""
```



Migration with Create migration

WHAT IS A MIGRATION?

A migration in Laravel is like version control for your database schema.

It lets you create, modify, or delete tables using PHP instead of raw SQL.



COMMAND TO CREATE A MIGRATION

```
php artisan make:migration create_products_table  
database/migrations/2025_10_08_0000_create_products_table.php
```



```
<?php
use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;
return new class extends Migration
{
    public function up(): void
    {
        Schema::create('products', function (Blueprint $table) {
            $table->id();
            $table->string('name');
            $table->text('description')->nullable();
            $table->decimal('price', 10, 2);
            $table->integer('stock')->default(0);
            $table->timestamps();
        });
    }
    public function down(): void
    {
        Schema::dropIfExists('products');
    }
};
```





THANK YOU

Do you have any questions?

02 OCTOBER,
2025